Model Evaluation Metrics

Understanding Key Performance Metrics: Accuracy, Precision, Recall, F1-Score, ROC-AUC & Confusion Matrix

Presenter

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Topics Covered

01

Confusion Matrix

02

Accuracy & Its Limits

03

Precision & Recall

04

F1-Score & ROC-AUC

05

Real-World Case Studies

MODULE 1 — FOUNDATIONS

Why Model Evaluation Matters

The Core Problem

A trained ML model is only as good as how well it generalizes to unseen data. Training accuracy alone is deeply misleading — a model can memorize training data perfectly while failing on new inputs.

Real-World Consequence

A spam filter with 95% accuracy but poor recall lets 50% of spam through — users suffer despite "high accuracy." High accuracy does not equal high utility.

What Evaluation Answers

→ **Correctness**

How often is the model correct overall?

→ **Reliability**

When it predicts positive, is it right?

→ **Completeness**

Does it catch all actual positives?

Medical Diagnosis

Minimize false negatives → missed disease

Fraud Detection

Minimize false positives → false alarms

Search Engines

Balance precision & recall simultaneously

Key Coverage

Confusion Matrix → Accuracy → Precision → Recall → F1 → ROC-AUC

MODULE 1 — FOUNDATIONS

Binary Classification: The Foundation

Definition & Examples

Binary classification assigns each input to one of two classes — **Positive (1)** or **Negative (0)**. The model outputs a probability score (0 to 1), converted to a class label using a threshold (default: 0.5).

- **Email:** Spam (1) or Not Spam (0)
- **Patient:** Disease (1) or Healthy (0)
- **Transaction:** Fraud (1) or Legitimate (0)

Four Possible Outcomes for Every Prediction

✅ True Positive (TP)

Predicted Positive, Actually Positive — correct identification

✅ True Negative (TN)

Predicted Negative, Actually Negative — correct rejection

❌ False Positive (FP)

Predicted Positive, Actually Negative — **Type I Error**

❌ False Negative (FN)

Predicted Negative, Actually Positive — **Type II Error**

These four values — TP, TN, FP, FN — form the **Confusion Matrix**, the single building block from which ALL evaluation metrics are derived.

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MODULE 2 — CONFUSION MATRIX

Confusion Matrix: Structure & Anatomy

A **2×2 table** comparing actual vs. predicted class labels for a classification model. Diagonal cells represent correct predictions; off-diagonal cells represent errors.

	Predicted: Positive (1)	**Predicted: Negative (0)**
Actual: Positive (1)	**TP (True Positive)**	FN (False Negative) ← **Type II Error**
Actual: Negative (0)	FP (False Positive) ← **Type I Error**	**TN (True Negative)**

Correct Predictions (Diagonal)

- **TP:** Model correctly identifies disease — patient is sick, model says sick
- **TN:** Model correctly clears healthy patient — patient is healthy, model says healthy

Errors (Off-Diagonal)

- **FP (Type I):** Healthy patient flagged as sick → unnecessary treatment, cost, anxiety
- **FN (Type II):** Sick patient cleared as healthy → missed diagnosis → most dangerous in medical context

Multi-class extension: For N classes, the confusion matrix becomes N×N. For example, a cat/dog/bird classifier uses a 3×3 matrix where each row represents the actual class and each column represents the predicted class.

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MODULE 2 — CONFUSION MATRIX

Confusion Matrix: Worked Example (Email Spam)

Scenario: 100 emails tested — 60 actual spam, 40 actual not-spam. Model predictions: TP = 50, FN = 10, FP = 5, TN = 35.

	Predicted: Spam	Predicted: Not Spam	Row Total
Actual: Spam	**TP = 50**	FN = 10	60
Actual: Not Spam	FP = 5	**TN = 35**	40
Column Total	55	45	**100**

Interpretation

- Model caught **50 of 60 spam emails** (missed 10)
- Flagged **5 legitimate emails** as spam
- **FN impact:** 10 spam emails reached inbox — user inconvenienced
- **FP impact:** 5 legitimate emails blocked — potentially critical emails lost

Key Insight

Raw accuracy = (50+35)/100 = **85%** — but this single number hides 10 missed spams and 5 false alarms. The confusion matrix reveals what accuracy conceals.

Next step: Derive Accuracy, Precision, Recall, and F1 from these four values.

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MODULE 3 — ACCURACY

Accuracy: Definition, Formula & Limitations

Formula & Calculation

Accuracy = (TP + TN) / (TP + TN + FP + FN)

Spam example: Accuracy = (50 + 35) / 100 = **85%**

Intuition: "Out of all predictions, how many were correct?" — the most intuitive and widely reported metric.

When Accuracy Works Well

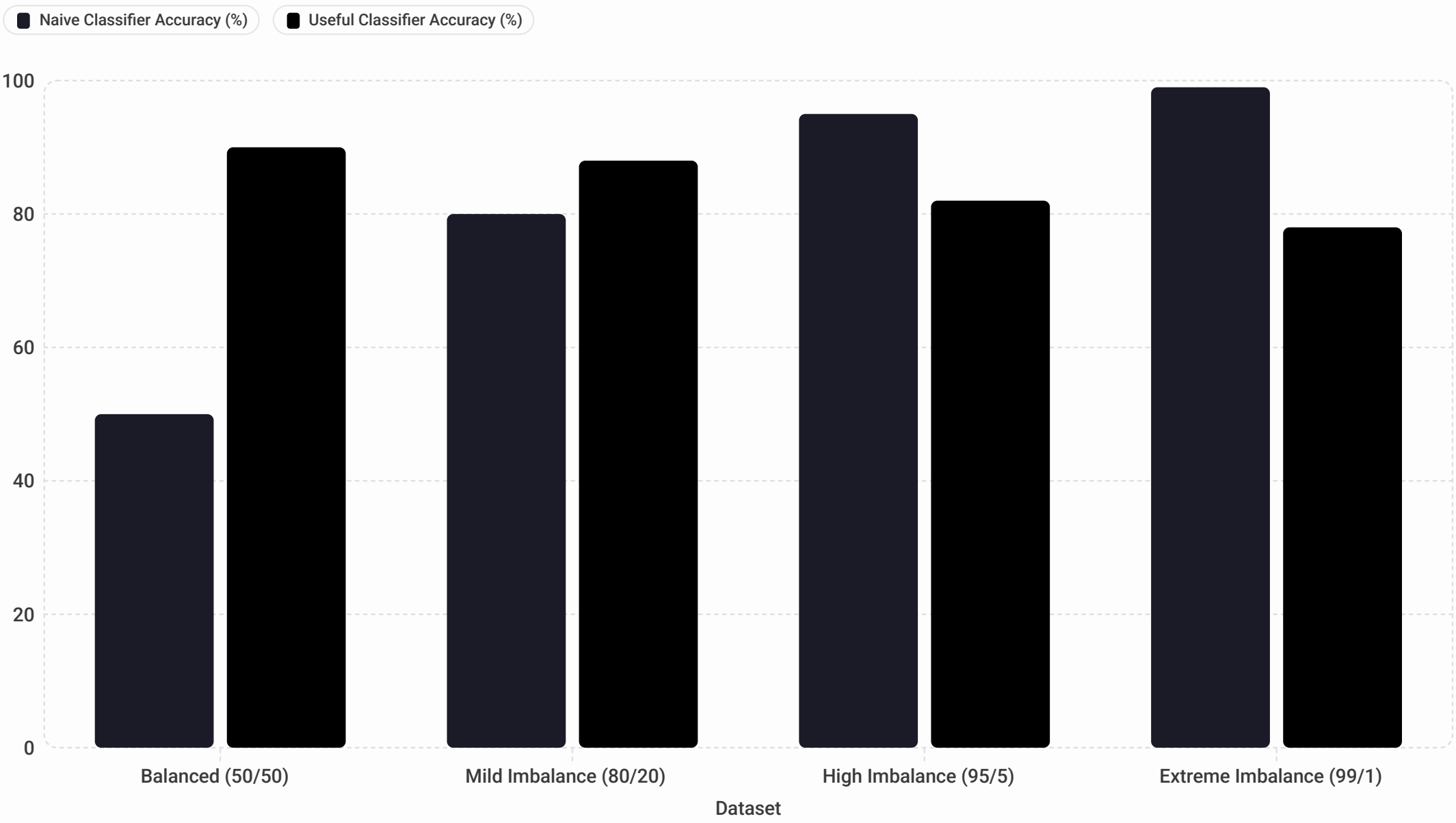
- Balanced datasets where both classes have roughly equal sample counts
- Simple classification tasks with low stakes errors

⚠️ The Accuracy Paradox — Class Imbalance

- Dataset: 950 healthy, 50 sick patients (95% healthy)
- A model that **always predicts "Healthy"** achieves **95% accuracy**
- Yet it catches **zero sick patients** — completely useless clinically
- Accuracy is deeply misleading when classes are imbalanced

Real-World Imbalanced Datasets

- Fraud detection (0.1% fraud transactions)
- Cancer detection (rare positive cases)
- Network intrusion (rare attacks vs. normal traffic)



As imbalance increases, a naive classifier that always predicts the majority class achieves deceptively high accuracy while a useful classifier (that actually detects minority cases) may score lower — **solution: use Precision, Recall, F1-Score, and ROC-AUC for imbalanced problems.**

MODULE 4 — PRECISION

Precision: Formula, Intuition & Use Cases

Formula & Calculation

Precision = TP / (TP + FP)

Spam example: Precision = 50 / (50 + 5) = 50/55 = **90.9%**

Intuition

"Of all emails the model flagged as spam, what % were actually spam?" — measures the **quality of positive predictions**, not their quantity.

- **High Precision:** When model says "Positive", it is almost always right — few false alarms
- **Low Precision:** Model raises many false alarms — many FP predictions pollute results

When to Prioritize Precision

📧 Spam Filtering

FP = legitimate email blocked → high cost → maximize precision above all

⚖️ Legal Document Review

FP = irrelevant document reviewed → wastes expensive lawyer time

🛒 Product Recommendations

FP = irrelevant recommendation → user frustration and disengagement

Precision-Recall Trade-off

Increasing the classification threshold → fewer positives predicted → **higher Precision but lower Recall**. The model becomes more selective.

⚠️ Limitation of Precision Alone

Precision ignores FN — a model that predicts only 1 positive (and gets it right) has **100% precision** but misses everything else. Never use precision in isolation.

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MODULE 5 — RECALL

Recall (Sensitivity): Formula, Intuition & Use Cases

Formula & Calculation

Recall = TP / (TP + FN)

Spam example: Recall = 50 / (50 + 10) = 50/60 = **83.3%**

Intuition

"Of all actual spam emails, what % did the model catch?" — measures the **completeness of positive detection**.

Also called: Sensitivity, True Positive Rate (TPR), Hit Rate

- **High Recall:** Catches most actual positives — few missed cases (low FN)
- **Low Recall:** Misses many actual positives — dangerous in critical applications

When to Prioritize Recall

🏥 Cancer / Disease Detection

FN = missed diagnosis → patient goes untreated → life-threatening consequence

💳 Fraud Detection

FN = missed fraud transaction → financial loss directly to the bank

🔒 Security Intrusion Detection

FN = missed attack → system compromised, data breach

Recall-Precision Trade-off

Lowering threshold → more positives predicted → **higher Recall but lower Precision** (more false alarms generated).

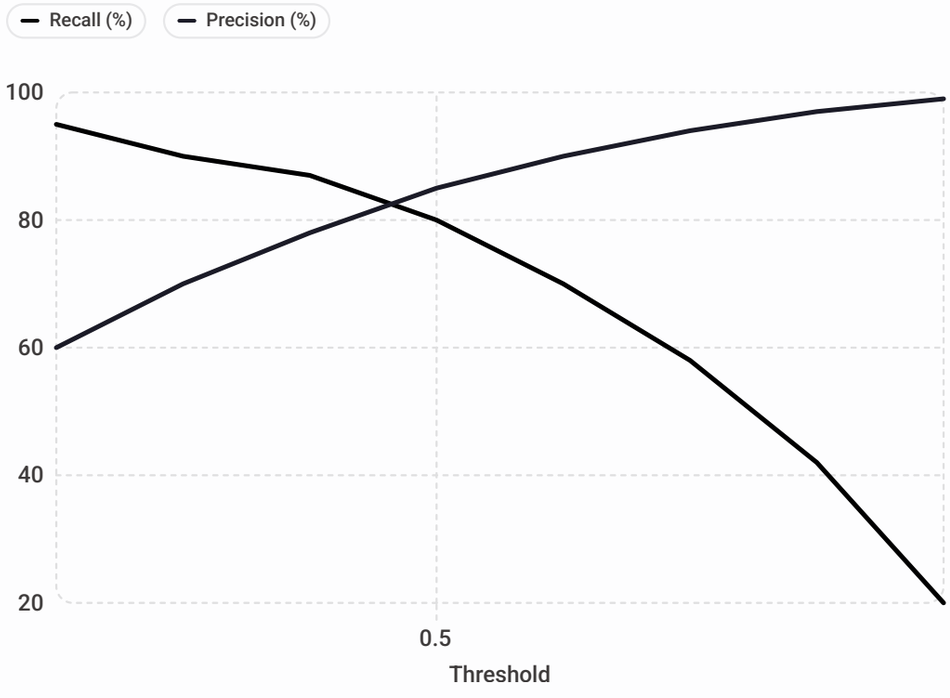
Specificity (Complement of FPR)

Specificity = TN / (TN + FP) — measures how well the model correctly identifies negative cases. High specificity = few false alarms.

MODULE 5 — THE TRADE-OFF

Precision vs. Recall: The Trade-off Explained

Core tension: Improving Precision often reduces Recall, and vice versa — this is controlled entirely by the classification threshold. Understanding this trade-off is essential for deploying models in real-world systems.



Threshold Effects

Threshold = 0.5 (Default)

Balanced starting point — equal weight to FP and FN costs

Threshold ↑ (e.g., 0.8)

Model is more conservative → fewer positives predicted → Precision ↑, Recall ↓

Threshold ↓ (e.g., 0.3)

Model is more aggressive → more positives predicted → Recall ↑, Precision ↓

Business Decision Rule

- **Medical:** Low threshold → High Recall priority (catch all sick)
- **Spam filter:** High threshold → High Precision priority (don't block legit mail)
- **Solution to the dilemma:** F1-Score — harmonic mean that balances both

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MODULE 6 — F1-SCORE

F1-Score: The Harmonic Balance

Formula

F1 = 2 × (Precision × Recall) / (Precision + Recall)

Spam example: F1 = 2 × (0.909 × 0.833) / (0.909 + 0.833) = 2 × 0.757 / 1.742 = **86.9%**

Why Harmonic Mean?

The arithmetic mean (P+R)/2 is misleading. Example: if P=1.0, R=0.01:

- Arithmetic mean = 0.505 (looks decent)
- Harmonic mean (F1) = **0.02 (reveals the truth)**

F1 is high only when **BOTH** Precision AND Recall are high — it severely punishes extreme imbalance between the two.

Generalization — Fβ Score

$F\beta = (1+\beta^2) \times (P \times R) / (\beta^2 \times P + R)$

- β = 0.5 → weights **Precision** more (spam filter)
- β = 1.0 → equal weight (standard F1)
- β = 2.0 → weights **Recall** more (medical diagnosis)

Interpretation

- **F1 = 1.0:** Perfect precision and recall
- **F1 = 0:** Either precision or recall is zero

When to Use F1

- Imbalanced datasets
- When both FP and FN costs matter equally

Limitation

F1 doesn't account for TN — use MCC (Matthews Correlation Coefficient) when TN also matters significantly.

Metric	Formula	Spam Example	Captures
Accuracy	(TP+TN)/(Total)	**85%**	Overall correctness
Precision	TP/(TP+FP)	**90.9%**	Quality of +ve predictions
Recall	TP/(TP+FN)	**83.3%**	Coverage of actual +ves
F1-Score	2×P×R/(P+R)	**86.9%**	Harmonic balance of P & R

ROC Curve: Concept & Construction

What is the ROC Curve?

ROC = Receiver Operating Characteristic — originated in WWII radar signal detection theory. It plots **True Positive Rate (Sensitivity/Recall)** on the Y-axis vs. **False Positive Rate (FPR)** on the X-axis, across ALL possible classification thresholds.

FPR Formula: FPR = FP / (FP + TN) = 1 − Specificity

Key Reference Points

- **Point (0,0):** Threshold = 1.0 → predict nothing as positive
- **Point (1,1):** Threshold = 0.0 → predict everything as positive
- **Point (0,1):** Perfect classifier — 100% TPR, 0% FPR
- **Diagonal (y=x):** Random classifier — no discrimination ability

Better model = curve closer to top-left corner

How to Construct the ROC Curve (Step-by-Step)

1 **Get Probability Scores**

Obtain model's probability scores for all test samples

2 **Sort Descending**

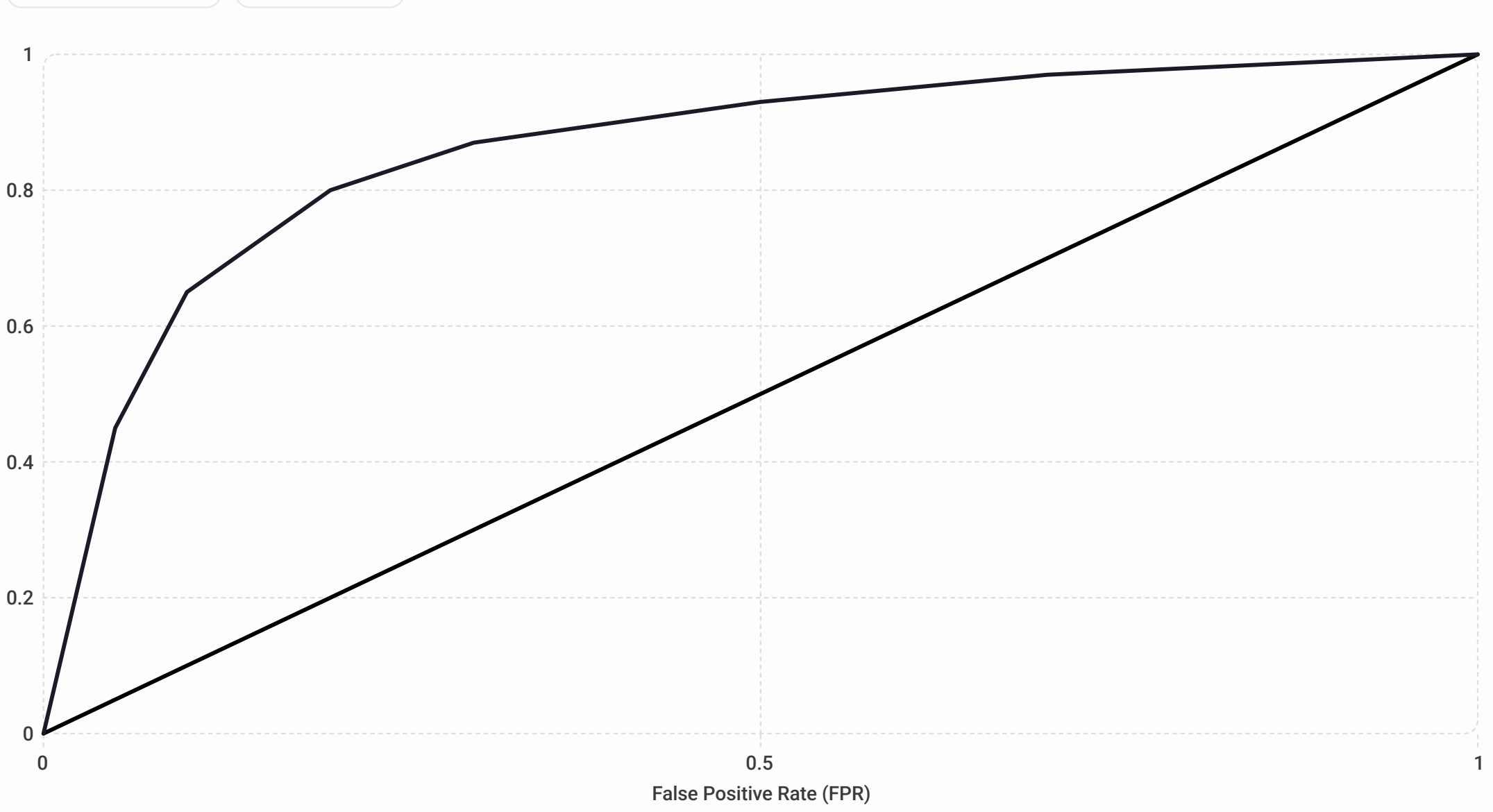
Sort samples by predicted probability from high to low

3 **Sweep Thresholds**

For each threshold from 1.0 → 0.0, compute TPR and FPR

4 **Plot & Connect**

Plot each (FPR, TPR) point and connect to form the curve



MODULE 7 — AUC

AUC: Area Under the ROC Curve

Definition & Interpretation

AUC = Area Under the ROC Curve — a single scalar value summarizing the entire ROC curve's performance across all thresholds.

Range: 0.0 to 1.0 (higher is better)

Probabilistic interpretation: AUC = probability that the model ranks a random positive sample higher than a random negative sample.

Key Advantages

- **Threshold-independent:** Evaluates model across all operating points simultaneously
- **Scale-invariant:** Measures ranking quality, not absolute probability scores

⚠️ Limitation

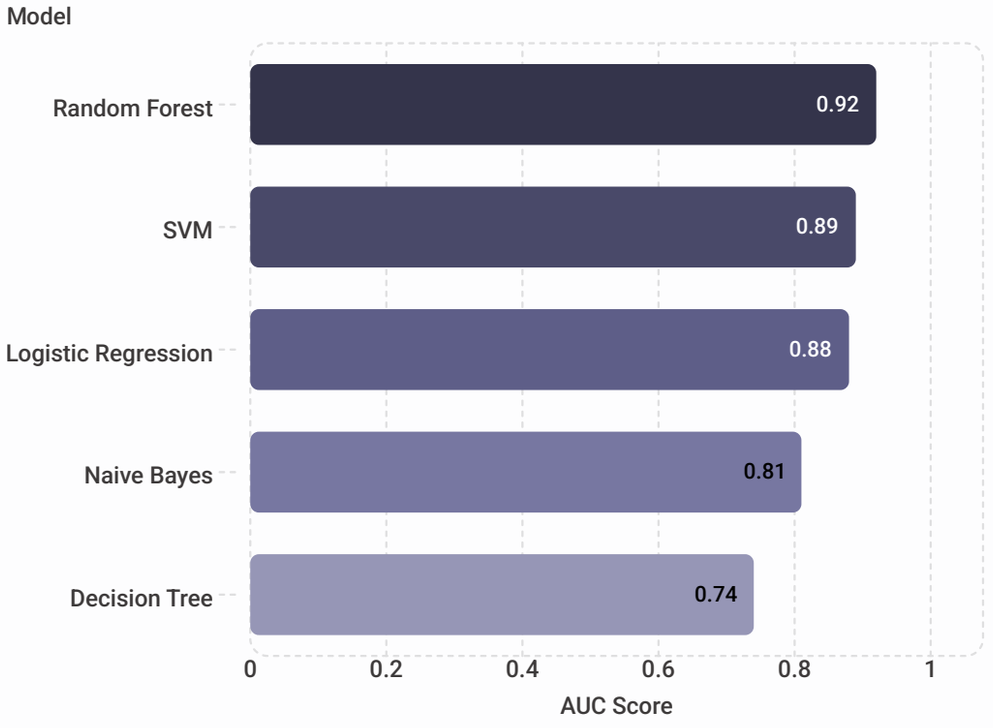
AUC can be misleading on highly imbalanced datasets → use **PR-AUC (Precision-Recall AUC)** instead for fraud, rare disease detection.

AUC Range	Model Quality	Interpretation
1.0	Perfect	Ideal — rarely achieved in practice (may indicate data leakage)
0.90 – 0.99	Excellent	Outstanding discrimination ability
0.80 – 0.89	Good	Strong model, suitable for most applications
0.70 – 0.79	Fair	Acceptable, room for improvement
0.60 – 0.69	Poor	Weak model, needs significant improvement
0.50	Random	No better than flipping a coin
< 0.50	Worse than random	Model predictions are inverted — flip the labels

MODULE 7 — MODEL COMPARISON

Comparing Multiple Models Using ROC-AUC

Primary use: Select the best model by comparing AUC values across classifiers trained on the same dataset and test split.



Analysis — Disease Detection Task

- **Winner: Random Forest (AUC = 0.92)** — best at ranking sick patients above healthy ones
- SVM (0.89) and Logistic Regression (0.88) are strong alternatives
- Decision Tree (0.74) is the weakest — likely overfitting without pruning

Threshold Selection from ROC

- **High sensitivity needed** (cancer screening) → pick point with TPR ≥ 0.95
- **High specificity needed** (confirmatory test) → pick point with FPR ≤ 0.05
- **Youden's Index:** Optimal threshold = argmax(TPR − FPR) — maximizes both sensitivity and specificity simultaneously

Cross-Validation Best Practice

Always report **mean AUC ± std deviation** across k-folds to ensure robustness and avoid overfitting to a single test split.

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MODULE 8 — MULTI-CLASS EVALUATION

Multi-Class Evaluation: Extending the Metrics

The Challenge

Binary metrics (TP, FP, FN, TN) must be extended for problems with 3 or more classes. A handwritten digit classifier (0–9) produces a 10×10 confusion matrix — 100 cells to interpret.

Three Averaging Strategies

Macro Average

Compute metric per class, take **unweighted mean** → treats all classes equally regardless of size. Best when all classes are equally important.

Weighted Average

Compute metric per class, **weight by class support** (sample count) → accounts for class imbalance. Recommended for imbalanced multi-class problems.

Micro Average

Aggregate all TP, FP, FN globally, then compute → **dominated by majority class**. Use only when majority class performance is the priority.

3-Class Fruit Classifier Example (Apple / Banana / Cherry)

Class	TP	FP	FN	Precision	Recall	F1	Support
Apple	45	5	8	0.90	0.85	0.87	53
Banana	38	7	4	0.84	0.90	0.87	42
Cherry	29	3	9	0.91	0.76	0.83	38
Macro Avg	—	—	—	**0.88**	**0.84**	**0.86**	—
Weighted Avg	—	—	—	**0.88**	**0.85**	**0.86**	133

Multi-Class ROC-AUC

Use **One-vs-Rest (OvR)** or **One-vs-One (OvO)** strategy; report macro/weighted AUC across all class pairs for a complete picture.

MODULE 9 — CASE STUDY 1

Real-World Case Study 1: Medical Diagnosis (Cancer Detection)

Problem: Classify mammogram images as Malignant (cancer) or Benign — 1,000 test samples: 100 malignant, 900 benign. **Model results:** TP=90, FN=10, FP=45, TN=855.

Metric	Value	Clinical Interpretation
Accuracy	(90+855)/1000 = **94.5%**	Misleading — 90% baseline by predicting all benign
Precision	90/(90+45) = **66.7%**	1 in 3 flagged patients is actually healthy (unnecessary biopsy)
Recall	90/(90+10) = **90%**	Catches 90 of 100 cancers — 10 missed diagnoses
F1-Score	2×0.667×0.9/(0.667+0.9) = **76.4%**	Balanced view of precision and recall together
AUC	**0.94**	Excellent discrimination ability across all thresholds

Clinical Decision

Recall is paramount — missing cancer (FN) is far more dangerous than an unnecessary biopsy (FP). The 10 missed diagnoses represent patients who receive no treatment.

Action: Lower threshold from 0.5 → 0.3 to boost Recall to **97%** (at cost of Precision dropping to 55%).

Lessons Learned

- Always align metric priority with the **cost of each error type** in your domain
- 94.5% accuracy is misleading — a "predict all benign" model scores 90%
- **Regulatory context:** FDA requires sensitivity ≥ 85% for AI-assisted cancer screening tools
- Threshold tuning is a clinical decision, not just a technical one

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MODULE 9 — CASE STUDY 2

Real-World Case Study 2: Credit Card Fraud Detection

Problem: Detect fraudulent transactions — 10,000 transactions: 100 fraud (1%), 9,900 legitimate. **Naive model (predict all legitimate):** Accuracy = 99% — but catches 0 fraud cases. **Trained model results:** TP=80, FN=20, FP=200, TN=9700.

Metric	Naive Model	Trained Model	Preferred?
Accuracy	99%	97.8%	❌ Misleading
Precision	N/A	80/(80+200) = **28.6%**	Higher = fewer false fraud alerts
Recall	0%	80/(80+20) = **80%**	Higher = catch more fraud
F1-Score	0	**0.42**	Trained model wins decisively
PR-AUC	~0.01	**0.71**	Trained model wins decisively

Business Trade-off

- **FP =** Customer card blocked — bad customer experience, potential churn
- **FN =** Fraud transaction goes through — direct financial loss to the bank
- **Strategy:** Use PR-AUC (not ROC-AUC) for imbalanced fraud data
- Optimize F1 or Fβ with **β=2** (recall-weighted) to penalize missed fraud

Industry Benchmark & Key Lesson

Visa/Mastercard target: Recall ≥ 85% with Precision ≥ 30% for real-time fraud detection systems.

⚠ Never use accuracy alone on imbalanced datasets — it creates a false sense of security and can mask completely broken models.

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MODULE 10 — DECISION FRAMEWORK

Choosing the Right Metric: Decision Framework

3-Step Decision Process

1 Is your dataset balanced or imbalanced?

Balanced → Accuracy is a valid starting metric. Imbalanced → Avoid accuracy; use F1, PR-AUC, or MCC.

2 What is the cost of each error type?

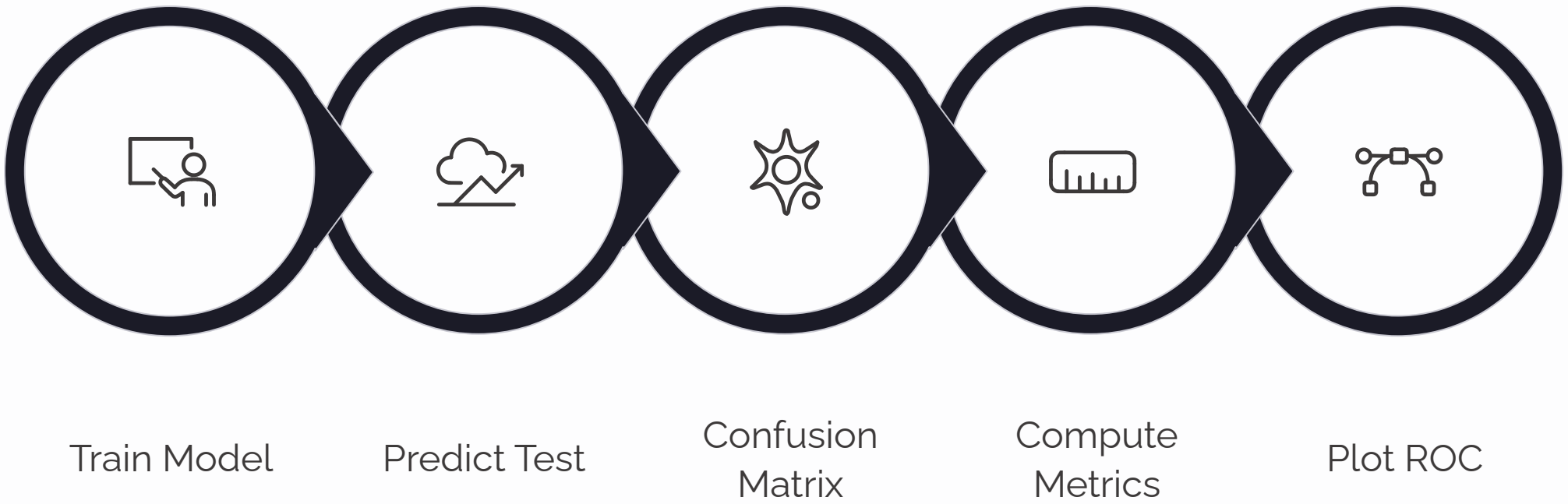
FN costly (missed disease, missed fraud) → Maximize Recall. FP costly (false alarm, blocked email) → Maximize Precision. Both matter equally → Maximize F1-Score.

3 Threshold-independent evaluation needed?

Yes → Use ROC-AUC (balanced data) or PR-AUC (imbalanced data).

Application	Primary Metric	Secondary Metric	Avoid
Medical Diagnosis	**Recall (Sensitivity)**	F1, AUC	Accuracy alone
Spam Filtering	**Precision**	F1	Recall alone
Fraud Detection	**Recall + PR-AUC**	F1 (β=2)	Accuracy
Search/Recommendation	**Precision@K**	NDCG	Raw Accuracy
Balanced Classification	**Accuracy + F1**	ROC-AUC	—
Multi-class (imbalanced)	**Weighted F1**	Macro AUC	Micro Accuracy

Complete Metrics Workflow: End-to-End Flow



Step	Phase	Key Output
1	Train Model (Logistic Regression, Random Forest, etc.)	Fitted model weights
2	Predict on held-out test set	Probability scores
3	Build Confusion Matrix at threshold 0.5	TP, TN, FP, FN values
4	Compute Metrics: Accuracy → Precision → Recall → F1	Accuracy, P, R, F1
5	Plot ROC Curve across all thresholds	AUC score
6	Select optimal threshold using Youden's Index or cost function	Optimal cutoff
7	Cross-validate with k-fold CV	Mean ± Std for all metrics
8	Compare models using AUC, F1, and domain metric	Best model selected
9	Deploy model; monitor metrics on live data (concept drift)	Live performance tracking

MODULE 12 — FORMULA REFERENCE

Quick Reference: All Formulas at a Glance

Metric	Formula	Range	Best Value	Key Question Answered
Accuracy	(TP+TN)/(TP+TN+FP+FN)	0–1	1.0	How often is the model correct overall?
Precision	TP/(TP+FP)	0–1	1.0	When model predicts +ve, how often is it right?
Recall / Sensitivity	TP/(TP+FN)	0–1	1.0	Of all actual +ves, how many did model catch?
Specificity	TN/(TN+FP)	0–1	1.0	Of all actual -ves, how many did model reject?
F1-Score	2×P×R/(P+R)	0–1	1.0	Harmonic balance of Precision & Recall
Fβ-Score	(1+β²)×P×R/(β²×P+R)	0–1	1.0	Weighted balance (β>1 favors Recall)
False Positive Rate	FP/(FP+TN)	0–1	0.0	How often does model raise false alarms?
ROC-AUC	Area under ROC curve	0–1	1.0	Overall ranking ability across all thresholds
PR-AUC	Area under P-R curve	0–1	1.0	Performance on imbalanced datasets

Golden Rules

- **No single metric tells the full story** — always report Confusion Matrix + F1 + AUC together
- **For imbalanced data:** PR-AUC > ROC-AUC > Accuracy
- **For balanced data:** Accuracy + ROC-AUC + F1 give a complete picture

Scikit-learn Functions

```
from sklearn.metrics import (
  confusion_matrix,
  classification_report,
  roc_auc_score,
  f1_score,
  precision_score,
  recall_score
)
```

Key Takeaways & Student Practice Problems

Core Concepts Summary

Confusion Matrix

The foundation — all metrics derive from TP, TN, FP, FN. Master this first.

Accuracy

Intuitive but dangerous on imbalanced datasets — always check class distribution first.

Precision vs. Recall

Quality vs. completeness — they trade off against each other via the threshold.

F1-Score

Harmonic mean that penalizes extreme imbalance between Precision and Recall.

ROC-AUC & PR-AUC

Threshold-independent ranking metrics; PR-AUC preferred for imbalanced problems.

Practice Problems for Students

1

Compute All 6 Metrics

Given TP=70, FP=30, FN=20, TN=80 → compute Accuracy, Precision, Recall, F1, Specificity, and FPR. Interpret each result.

2

Plot the ROC Curve

For a binary classifier with 10 sample predictions (provided scores and labels), compute TPR/FPR at each threshold and plot the ROC curve. Calculate AUC manually.

3

Explain the Paradox

Explain with a numeric example why a 99% accurate model can be completely useless. What metric should be used instead?

Domain Context Determines Metric Priority

- **Medical** → Recall (minimize missed diagnoses)
- **Spam** → Precision (minimize blocked legitimate mail)
- **Fraud** → F1 + PR-AUC (balance, imbalanced data)

Further Reading

Scikit-learn documentation: `classification_report()`, `roc_auc_score()`, `confusion_matrix()`

APPENDIX — EXTENDED COVERAGE

The Accuracy Paradox: A Detailed Walkthrough

Scenario Setup

Consider a **rare disease dataset**: 950 healthy patients and 50 sick patients in a 1,000-sample test set. The disease prevalence is just 5%.

The Lazy Model

A model that simply predicts "Healthy" for every single patient achieves:

- TP = 0 (catches no sick patients)
- TN = 950 (correctly identifies all healthy)
- FP = 0 (never raises false alarms)
- FN = 50 (misses every sick patient)
- **Accuracy = 950/1000 = 95%** ← misleading!

Why the Paradox Occurs

Accuracy rewards the model for correctly predicting the **majority class**. When one class dominates, the denominator (total samples) is driven by the majority, inflating the score even when minority class performance is zero.

Precision = 0/0

Undefined — model never predicts positive

Recall = 0/50 = 0%

Zero sensitivity — all sick patients missed

F1 = 0

Exposes the failure that accuracy hides

This is why F1-Score, Recall, and PR-AUC are mandatory for any imbalanced classification task in production systems.

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APPENDIX — THRESHOLD TUNING

Threshold Selection: Balancing Domain Requirements

The default classification threshold of 0.5 is rarely optimal in real-world deployments. Threshold selection is a **business and clinical decision** as much as a technical one.

Youden's Index

J = TPR − FPR = Sensitivity + Specificity − 1

The optimal threshold is the point on the ROC curve that **maximizes J** — simultaneously maximizing sensitivity and specificity. It represents the maximum vertical distance from the diagonal random-chance line.

- J = 0: No better than random
- J = 1: Perfect discrimination
- Youden's Index is scale-invariant and threshold-agnostic

Domain-Specific Threshold Rules

Cancer Screening

Require TPR ≥ 0.95 → accept lower threshold → tolerate more false positives to catch all cancer cases

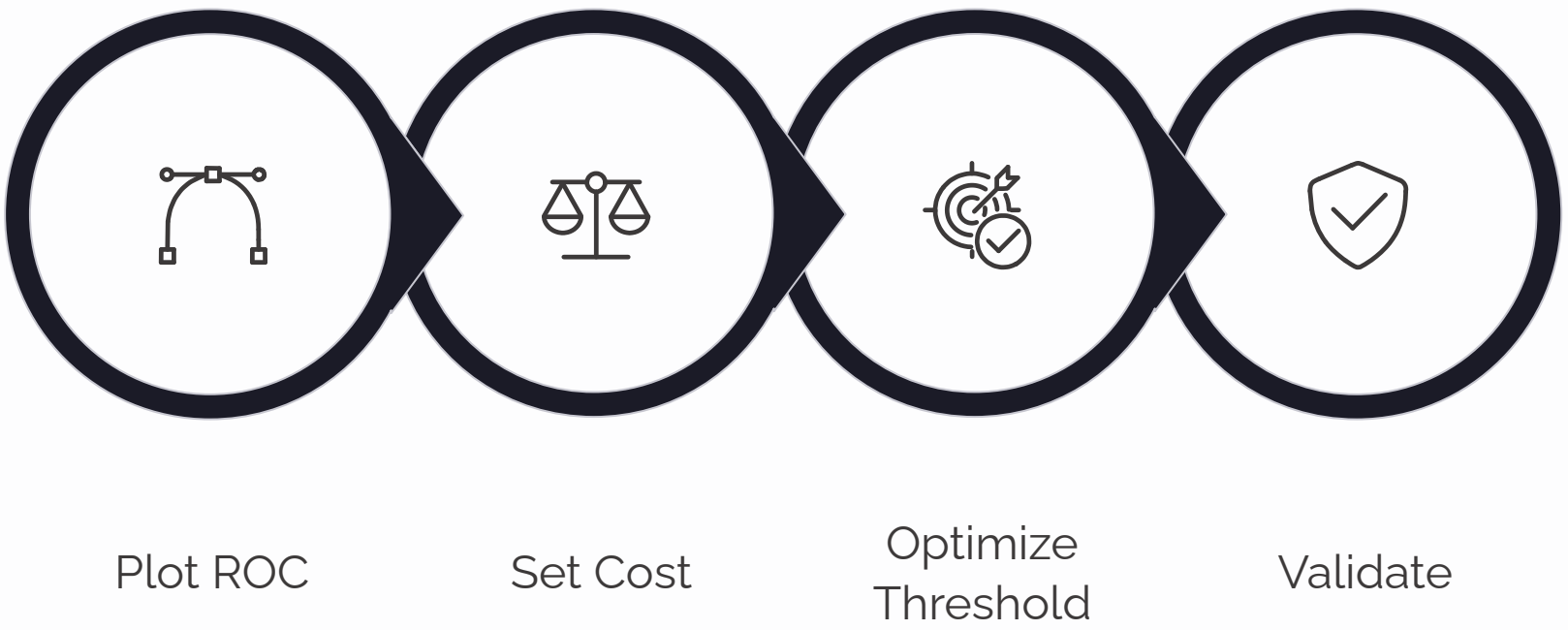
Confirmatory Testing

Require FPR ≤ 0.05 → accept higher threshold → very specific, fewer false alarms

Cost-Sensitive Selection

Minimize: C(FN)×FN_rate + C(FP)×FP_rate where C() is the domain cost of each error type

Practical Threshold Tuning Process



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APPENDIX — PR CURVE

Precision-Recall Curve & PR-AUC

What is the PR Curve?

The **Precision-Recall Curve** plots Precision on the Y-axis against Recall on the X-axis across all classification thresholds. It is the preferred evaluation curve for **highly imbalanced datasets** where the positive class is rare.

Why Not Just Use ROC-AUC?

ROC-AUC uses TN in the FPR calculation. When TN is very large (millions of legitimate transactions), FPR stays artificially low — making a poor model look good. PR-AUC focuses exclusively on the **positive class performance**.

Baseline Comparison

- **ROC baseline:** Random classifier AUC = 0.5 (diagonal)
- **PR baseline:** Random classifier PR-AUC = prevalence rate (e.g., 0.01 for 1% fraud)
- A PR-AUC of 0.71 on 1% fraud data represents an **enormous** improvement over the 0.01 baseline

Interpreting the PR Curve Shape

Top-Right Bulge

Excellent model — high precision at high recall. Ideal shape bows toward the top-right corner.

Steeply Falling Curve

Model precision collapses quickly as recall increases — common in imbalanced settings.

Flat Horizontal Line

Model maintains precision across recall range — rare but ideal for fraud detection.

When to Use PR-AUC vs. ROC-AUC

Scenario	Preferred Metric
Balanced dataset	ROC-AUC
Imbalanced (fraud, rare disease)	PR-AUC
TN doesn't matter (screening)	PR-AUC
Full operating range needed	Both — report together

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APPENDIX — ADVANCED METRICS

Advanced Metrics: MCC, Cohen's Kappa & Log-Loss

Matthews Correlation Coefficient (MCC)

MCC = (TP×TN − FP×FN) / √((TP+FP)(TP+FN)(TN+FP)(TN+FN))

MCC accounts for ALL four cells of the confusion matrix — including TN. It is considered one of the most informative single-number metrics for binary classification.

- **Range:** −1 to +1
- **MCC = +1:** Perfect prediction
- **MCC = 0:** No better than random
- **MCC = −1:** Complete disagreement with labels
- **Best for:** Highly imbalanced datasets where F1 is insufficient

Cohen's Kappa (κ)

κ = (Observed Accuracy − Expected Accuracy) / (1 − Expected Accuracy)

Measures agreement between model predictions and true labels, adjusted for chance agreement. Useful for multi-class evaluation.

- κ > 0.8: Strong agreement
- κ = 0.6–0.8: Moderate agreement
- κ < 0.4: Weak agreement

Log-Loss (Cross-Entropy Loss)

Log-Loss = −(1/N) × Σ [y×log(p) + (1−y)×log(1−p)]

Measures the quality of probability calibration — not just which class is predicted, but how confident the model is. **Lower is better.** A perfectly calibrated model achieves Log-Loss = 0.

MCC is recommended by many ML practitioners as the single best metric for binary classification on imbalanced datasets, as it produces a high score only when predictions are good across all four confusion matrix cells.

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APPENDIX — SCIKIT-LEARN IMPLEMENTATION

Implementing Evaluation Metrics in Python (Scikit-learn)

Complete Evaluation Pipeline

```
from sklearn.metrics import (
    confusion_matrix,
    classification_report,
    roc_auc_score,
    roc_curve,
    f1_score,
    precision_score,
    recall_score,
    average_precision_score
)
import matplotlib.pyplot as plt

# 1. Train and predict
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
y_prob = model.predict_proba(X_test)[:, 1]

# 2. Confusion matrix
cm = confusion_matrix(y_test, y_pred)
print(cm)

# 3. Full classification report
print(classification_report(y_test, y_pred))

# 4. ROC-AUC
auc = roc_auc_score(y_test, y_prob)
fpr, tpr, thresholds = roc_curve(y_test, y_prob)

# 5. PR-AUC
pr_auc = average_precision_score(y_test, y_prob)

# 6. Youden's Index for optimal threshold
j_scores = tpr - fpr
optimal_idx = j_scores.argmax()
optimal_threshold = thresholds[optimal_idx]
print(f"Optimal threshold: {optimal_threshold:.3f}")
```

classification_report() Output

```
              precision  recall  f1-score  support

           0      0.92    0.97    0.94      855
           1      0.67    0.90    0.76      100

    accuracy                       0.94      955
   macro avg      0.79    0.93    0.85      955
weighted avg      0.89    0.94    0.92      955
```

Key Parameters to Know

- average='macro' — unweighted mean across